

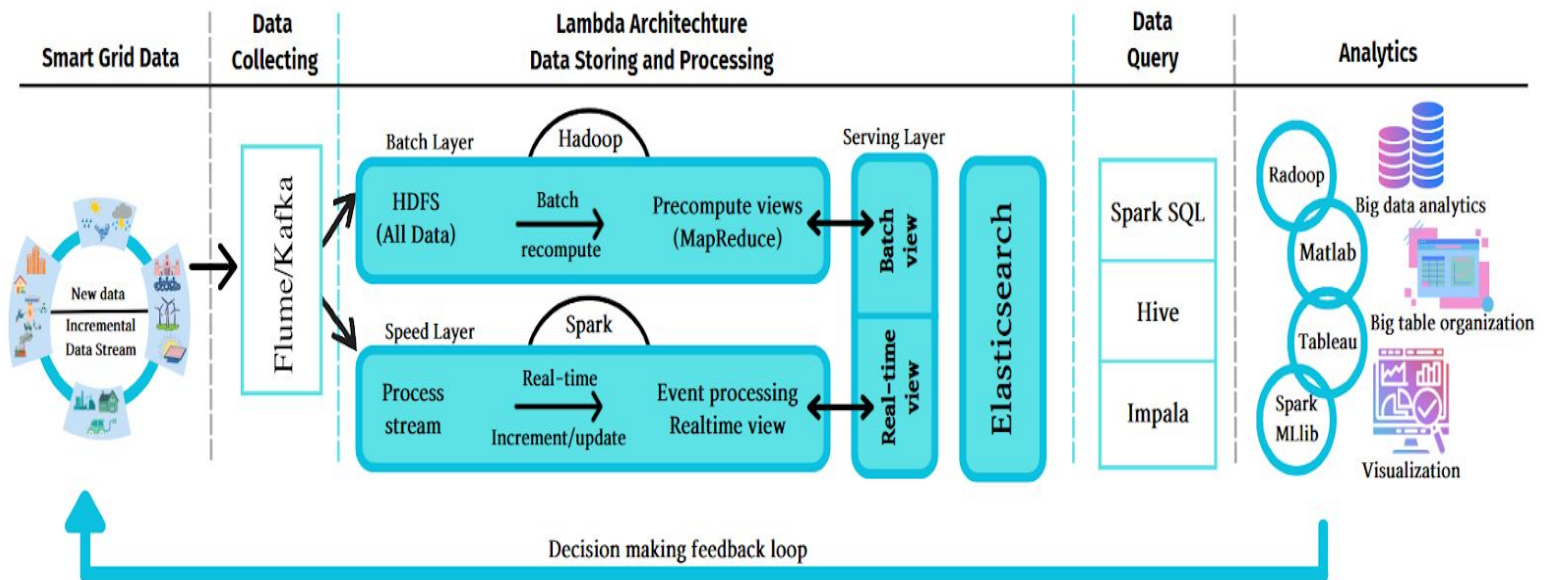


Project Big Data Storage and Processing

ONE LOVE. ONE FUTURE.

Objective

- Build a big data storage and processing architecture with a full range of components as shown below including : data collection, storage, analysis and visualization



Requirements

Direction 1: Building and install a big data architecture

- Do
 - Dashboard with charts that are helpful for understanding an overview of the data analyze
 - Data analysis: exploring the knowledge/information hidden under the data, helping to solve practical problems.
 - Environment: k8s or cloud (No docker)
 - Emulate distributed environment
 - Test scalability, fault-tolerance
- Don't
 - This course is not about Machine learning modeling

- Direction 2:
 - Research in deep a recent big data technology
 - Apache Iceberg, Druid, Deltalake, etc
 - Research in deep a big data related paper

Some directions

- **(Very simple, not to be complicated)**
 - Trend analysis
 - Detect unusual/typical events
 - Forecast price fluctuations
 - Detect fakes and scams
 - Evaluate KOLs on social networks
 - Analyze sentiments
 - Evaluate/forecast potential products

Project Timeline

+ 1W due to the delay

- Oct 2, 2024: List of groups (4-5 students/group) (file project.xlsx)
- Oct 3, 2024: creating channel (on MS. Teams) for each group
- Oct 6, 2024: Project description submission
- Oct 10, 2024: Feedback
- Nov 1, 2024: progress report
- Dec 9-25, 2024: Final presentation

(Project communication via group channel)

Project description

- Title:
 - E.g: apartment rental price analysis in Vietnam
- Data sources
 - E.g: Crawl from internet (list url)
- Requirements/Functions
 - Collecting data: describe data to be collected including key features such as apartment area, location, number of rooms, price..., identify problems/challenges (missing data, irregular data structure, heterogeneity...)
 - Storage requirement: expected volume of data for storage
 - Processing/analysis: list the various types of analysis and visualization requirement. E.g calculating the average price over a period, predicting the rental price
(statistical calculations should be included, and the use of machine learning/AI models is encouraged to discover hidden information within the data and assist in solving the practical problems identified)